

CENG381 Stochastic Processes

De Moivre's Martingale — Practice 1 (Answer Key)

Q1 Solution

a) Compensated walk is a martingale:

$$Y_n = Y_{n-1} + X_n - (p-q)$$

$$\begin{aligned} E[Y_n \mid Y_{n-1}, \dots, Y_1] &= Y_{n-1} + E[X_n - (p-q)] \\ &= Y_{n-1} + E[X_n] - (p-q) \\ &= Y_{n-1} + (p-q) - (p-q) \\ &= Y_{n-1} \quad \checkmark \end{aligned}$$

b) Expected values:

$$\begin{aligned} E[S_n \mid S_0=0] &= n(p-q) \quad (\text{drift } n(p-q) \text{ per step}) \\ E[Y_n] &= E[S_n - n(p-q)] = n(p-q) - n(p-q) = 0 \end{aligned}$$

Y_n is S_n with its expected trend removed. A martingale has constant expectation; subtracting the drift achieves this.

c) Numerical example with $p = 0.7$:

$$\mu = p - q = 0.7 - 0.3 = 0.4$$

$$(i) \quad \text{Drift per step} = 0.4$$

$$(ii) \quad S_0 = 5 \Rightarrow Y_n = S_n - 5 - n \cdot 0.4 \quad (\text{using } Y_0 = S_0 - 0 \cdot (p-q) = 5)$$

Actually $Y_n = S_n - n(p-q)$, so $Y_0 = S_0 = 5$ if $S_0=5$, and $Y_n = S_n - n \cdot 0.4$.

$$E[S_{100} \mid S_0=5] = 5 + 100 \cdot 0.4 = 45$$

Q2 Solution

a) Proof that $\{M_n = (q/p)^{S_n}\}$ is a martingale:

$$M_n = (q/p)^{S_{n-1} + X_n} = (q/p)^{S_{n-1}} * (q/p)^{X_n} = M_{n-1} * (q/p)^{X_n}$$

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$$\begin{aligned}E[(q/p)^{X_n}] &= p * (q/p)^1 + q * (q/p)^{-1} \\&= p * (q/p) + q * (p/q) \\&= q + p \\&= 1\end{aligned}$$

$$E[M_n | M_{n-1}, \dots] = M_{n-1} * E[(q/p)^{X_n}] = M_{n-1} * 1 = M_{n-1} \quad \checkmark$$

b) Gambler's Ruin formula via OST:

$$E[M_J] = M_0 = (q/p)^k$$

$$\text{At stopping: } M_J = (q/p)^N \text{ w.p. } p_k, \quad M_J = (q/p)^0 = 1 \text{ w.p. } 1-p_k$$

$$p_k * (q/p)^N + (1-p_k) * 1 = (q/p)^k$$

$$p_k * [(q/p)^N - 1] = (q/p)^k - 1$$

$$p_k = [(q/p)^k - 1] / [(q/p)^N - 1]$$

c) Limiting case $p = 1/2$ ($r = q/p \rightarrow 1$):

Let $r = q/p$. As $r \rightarrow 1$, both numerator $r^k - 1$ and denominator $r^N - 1 \rightarrow 0$.

$$\text{L'Hopital: } d/dr[r^k - 1] = k*r^{k-1}$$

$$d/dr[r^N - 1] = N*r^{N-1}$$

$$\lim_{r \rightarrow 1} (r^k - 1)/(r^N - 1) = k*1^{k-1} / (N*1^{N-1}) = k/N$$

So $p_k = k/N$ for $p=1/2$, matching the fair-martingale OST result. $\checkmark$

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Q3 Solution

a) Favorable game: $p=2/3$, $k=2$, $N=4$:

$$q/p = (1/3)/(2/3) = 1/2$$

$$\begin{aligned} p_2 &= [(1/2)^2 - 1] / [(1/2)^4 - 1] \\ &= [1/4 - 1] / [1/16 - 1] \\ &= [-3/4] / [-15/16] \\ &= (3/4) * (16/15) \\ &= 48/60 = 4/5 = 0.80 \end{aligned}$$

Sanity check: $p_2 = 0.80 > 0.50 \quad \checkmark$ (favorable game)

b) Unfavorable game: $p=1/3$, $k=3$, $N=5$:

$$q/p = (2/3)/(1/3) = 2$$

$$\begin{aligned} p_3 &= [2^3 - 1] / [2^5 - 1] \\ &= [8 - 1] / [32 - 1] \\ &= 7/31 \approx 0.226 \end{aligned}$$

Sanity check: $p_3 = 7/31 < 1/2 \quad \checkmark$ (unfavorable game)

c) Comparing p_1 , p_3 , p_4 (unfavorable game, $q/p=2$, $N=5$):

$$\begin{aligned} p_1 &= [2^1 - 1] / [2^5 - 1] = 1/31 \approx 0.032 \\ p_3 &= [2^3 - 1] / [2^5 - 1] = 7/31 \approx 0.226 \\ p_4 &= [2^4 - 1] / [2^5 - 1] = 15/31 \approx 0.484 \\ p_1 &< p_3 < p_4 < 1/2 \end{aligned}$$

Even starting with \$4 out of \$5, the probability of reaching the goal is only 48.4% when $p=1/3$. The function p_k is increasing in k (more starting capital helps even in an unfavorable game), but always stays below $1/2$ when $p < 1/2$.